

Autonomous Pharmaceutical Operations Intelligence

Data engineering + machine learning + qualitative NLP + constrained optimization

Project mission. Develop an auditable pharmaceutical operations decision-support system that integrates quantitative and qualitative public data to identify supply-risk signals, characterize demand pressure, and recommend resource-allocation strategies under explicit operational constraints. The system is designed to support - not replace - human decision makers.

138,217

FDA product
records

256,383

FDA package
records

17,975

FDA recall
events

1,601

FDA shortage
records

762,907

package-year
model rows

System architecture

1. INGEST	2. LINK	3. MODEL	4. INTERPRET	5. OPTIMIZE	6. EXPLAIN
FDA shortages, NDC + recalls CMS Part D	Python + MySQL validation + relational joins	Rare-event shortage-risk ranking	Rule taxonomy + NLP for recall language	OR-Tools constrained allocation	Human-reviewable outputs + limitations

Data sources

FDA Drug Shortages: 1,601 shortage rows used to identify package-level shortage events and historical recurrence signals.

FDA NDC Directory: 138,217 unique source products keyed by FDA **product_id** and 256,383 deduplicated package NDCs. Product NDC alone was not treated as a unique key.

FDA Recall / Enforcement: 17,975 recall events plus 25,577 valid recall-package relationships used for historical quality and supply-risk signals.

CMS Medicare Part D: 2021-2024 geography-and-drug utilization used as a demand-pressure proxy. Part D utilization is explicitly not treated as total U.S. pharmaceutical demand.

Integration results

Shortage-to-NDC: 1,570 unique shortage NDCs; 1,393 unique linked NDCs; **88.73% match rate**. The 180 unlinked shortage rows were preserved rather than forced into false matches.

Recall bridge: 11,233 unique recall package NDCs in the source; 8,606 unique linked package NDCs after relational matching. Missing package NDCs remain represented at the recall-event level.

Temporal panel: 762,907 package-year observations across 2022-2025 with only 426 shortage positives, creating an extreme rare-event prediction problem.

Temporal alignment: prior-year CMS utilization is aligned to the following shortage cohort (2021->2022, 2022->2023, 2023->2024, 2024->2025).

Technology stack: Python (pandas, scikit-learn) | MySQL | JupyterLab | OR-Tools | Excel/Tableau | Git/GitHub | FDA/openFDA public data | CMS public utilization data

What PharmaOpt has demonstrated so far

The project is intentionally evaluated as a sequence of testable analytical components rather than a single black-box model. Each component is retained, rejected, or bounded based on observed evidence.

1 | Shortage-risk modeling

Version	Main change	2025 development-holdout result	Decision
v0.1	Product age + recall history + product characteristics	Established a rare-event ranking baseline	Retained as baseline
v0.2	Added historical shortage recurrence features	Improved some top-k capture	Retained for comparison
v0.3	Added supply-pressure, active breadth, recency and sibling signals	PR-AUC 0.003345; 7.72x lift; ROC-AUC 0.7988	Best FDA-only version; stop FDA-only tuning
v0.4	Added CMS Part D utilization / demand-pressure features	PR-AUC 0.003133; 7.23x lift; ROC-AUC 0.7769 ; top 500 captured 10 positives	Useful top-ranking behavior; no over-tuning

Evaluation principle: shortages are extremely rare, so accuracy is not a useful headline metric. PharmaOpt emphasizes PR-AUC, lift over the positive-class baseline, ROC-AUC as a secondary metric, and top-k capture. The 2025 cohort is a **development holdout**, not a pristine final test set, because it has been inspected repeatedly during feature development.

2 | Qualitative supply-risk intelligence (Notebook 09)

Interpretable taxonomy v0.2

FDA recall reasons were normalized and mapped to multi-label operational-risk categories including sterility, manufacturing/GMP, contamination, particulate matter, potency/dosage, specification failure, stability/expiry, packaging/delivery, labeling, regulatory approval, storage temperature, product identity/formulation, and physical product quality.

Coverage: 17,545 of 17,975 recall records classified; 430 residual records remained. Coverage is approximately **97.61%** and is not described as accuracy because no independent ground-truth label set exists.

NLP validation and residual review

TF-IDF + K-means was tested on the 72 unique residual descriptions. Best silhouette = **0.0747** at k=10, so cluster membership was rejected as too weak for an operational feature.

A leakage-resistant weakly supervised multi-label classifier used word + character TF-IDF. Holdout: **Micro Precision 0.9939, Micro Recall 0.9755, Micro F1 0.9846, Macro F1 0.9432, Hamming Loss 0.0027**. These metrics measure generalization of the rule-derived taxonomy, not independent causal truth.

At a conservative review score ≥ 0.85 , NLP surfaced **9 unique residual descriptions representing 31 recall records**. Most residuals were intentionally left unresolved rather than force-classified.

3 | Resource-allocation optimization (Notebook 08)

Policy	Service range	Weighted unmet objective	Improvement vs proportional
Proportional baseline	70% everywhere	212,128.34	0.00%
Demand-priority	60%-100%	161,310.41	23.96%
Balanced PharmaOpt	60%-80%	180,587.88	14.87%

Case study: lisdexamfetamine dimesylate using 2024 CMS state-level utilization as an ingredient-level demand proxy. Scenario supply was deliberately modeled at **70% of demand** (304,247.65 supply vs 434,639.50 demand; 30% modeled deficit). This is not observed real inventory. The balanced policy used 60%-80% service bounds to reduce extreme geographic allocation differences.

Robustness: across modeled supply ratios from 50% to 90%, the balanced optimization improved the project's demand-weighted unmet-demand objective by **8.92% to 44.61%** versus proportional allocation. Total unmet demand is fixed by the modeled supply level; the optimization changes where that unmet demand falls according to the chosen objective and policy constraints.

Technical map, current status, and next phase

A compact technical reference for recruiters, professors, analysts, and technical reviewers.

Relational database design

Table	Purpose / key design choice
drug_product	138,217 products; FDA product_id is the primary key. product_ndc is indexed but not assumed unique.
drug_package	256,383 deduplicated package NDCs linked to product_id.
shortage_event	1,601 shortage records with original source NDC preserved and nullable linked package NDC.
recall_event	17,975 recall events keyed by recall_number; preserves qualitative reason and distribution text.
recall_package	25,577 valid recall-package relationships; bridge supports one recall affecting many packages.

Completed analytical workflow

Notebook	Purpose	Status
01	FDA shortage audit	Complete
02	FDA NDC audit	Complete
03	FDA recall audit	Complete
04	FDA data integration	Complete
05	MySQL data load / relational schema	Complete / frozen
06	Shortage-risk analytics + v0.1-v0.4 modeling	Complete through current model
07	CMS Part D utilization integration	Complete
08	Resource-allocation optimization + robustness	Complete / frozen
09	Qualitative supply-risk intelligence + NLP	Complete / frozen
10	Qualitative risk model integration	Next phase

Current project status

Notebook 09 is complete and frozen. PharmaOpt now has relational data infrastructure, rare-event shortage-risk models, CMS demand-pressure integration, an optimization prototype, and an auditable qualitative/NLP intelligence layer.

The project remains **in progress**. Existing results are research prototypes and portfolio evidence, not production deployment claims.

Next: model v0.5

Notebook 10 will create temporally valid features such as prior GMP-risk, sterility-risk, contamination-risk, qualitative-signal recency, and risk-category diversity before each cohort date.

Those features will be added to the FDA + CMS package-year panel and compared against v0.4 to test whether qualitative intelligence adds predictive value.

Methodological guardrails

• No causal claim: recall history, qualitative categories, utilization, and shortage outcomes may be associated without being causal.	• Rare-event problem: only 426 positives exist in the 2022-2025 package-year panel; accuracy can be misleading.
• Development holdout: 2025 has been inspected during feature development and is not a pristine final test set.	• NDC snapshot bias: current FDA NDC data may create survivorship / historical availability limitations.
• CMS scope: Part D utilization is a demand proxy, not total U.S. demand; ingredient-level utilization may not match a specific formulation.	• NLP scope: taxonomy coverage is not ground-truth accuracy; weakly supervised NLP reproduces rule-derived labels and unresolved text can remain unresolved.
• Optimization scope: supply ratios, service bounds, and objective weights are modeled assumptions; weighted unmet demand is not clinical harm.	• Human approval: consequential pharmaceutical distribution decisions should remain subject to qualified human review and operational governance.

How to explain PharmaOpt in 30 seconds

"PharmaOpt is an independent pharmaceutical operations intelligence project I am building to combine FDA shortage, NDC, recall, and CMS utilization data in a relational database. I use Python and machine learning to rank shortage risk, NLP to turn recall text into auditable operational-risk signals, and OR-Tools to test constrained allocation strategies when modeled supply is limited. The goal is a transparent decision-support system that helps analysts identify risk earlier, understand why a signal was raised, and compare allocation policies under explicit assumptions."

Joaquin Pizano
Artificial Intelligence & Business Analytics
University of South Florida

Project: PharmaOpt
GitHub: github.com/joaquinpizano/pharmaopt
Email: joaquinpizano@usf.edu